



1 Methodology

Biofilm biomass was quantified by crystal violet staining (OD570) and normalized to bacterial growth (OD600). Specific Biofilm Formation (SBF) Index was calculated as: $SBF = (OD570 \text{ sample} - OD570 \text{ blank}) / (OD600 \text{ sample} - OD600 \text{ blank})$. Wells were grouped by strain/condition and media condition, with blank subtraction performed per matching media condition.

Classification thresholds applied — Weak biofilm: $SBF \geq 0.5$; Moderate biofilm: $SBF \geq 0.75$; Strong biofilm: $SBF \geq 1.1$; below the weak threshold, wells were classed Non-biofilm.

2 Instrument & acquisition details

OD600 file — Device: — Wavelength: nm Date: 08 May 2026 Time: 11:02:40 AM Temp: —

OD570 file — Device: — Wavelength: nm Date: 08 May 2026 Time: 12:15:51 PM Temp: —

3 Summary statistics

Strains/Conditions analysed: 30 Media conditions: 2 Blank wells: 6

Strong biofilm: 7 Moderate biofilm: 2 Weak biofilm: 0 Non-biofilm: 21

4 SBF Index by strain/condition — Figure 1



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Specific Biofilm Formation · crystal violet (OD570) normalized to growth (OD600)

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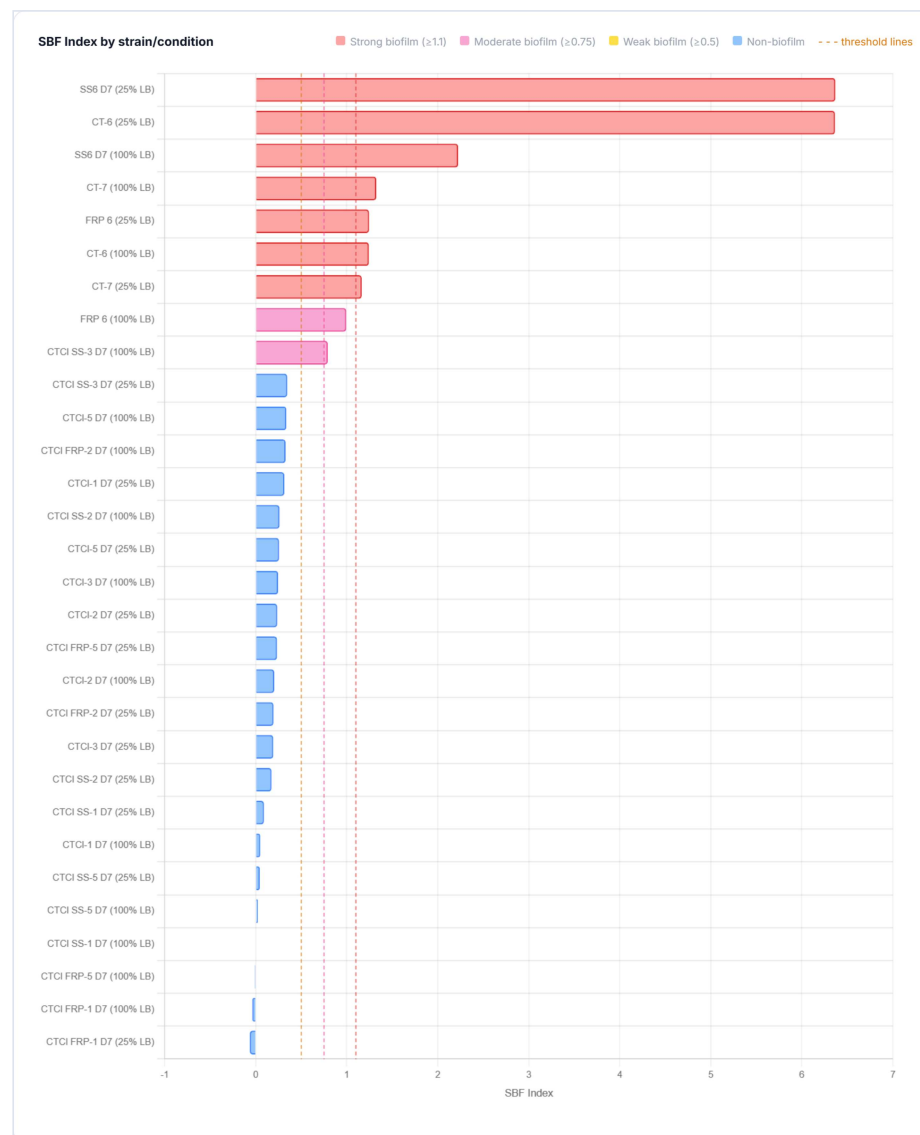




Figure 1. SBF Index per strain/condition, colour-coded by classification. Dashed lines mark the weak/moderate/strong thresholds.

5 Detailed results table — Table 1

Strain/Condition / ID	Type	Media	n	Mean OD600	Mean OD570	Blank OD600	Blank OD570	SBF Index	% Inhib.	Classification
SS6 D7	Strain/Condition	25% LB	3	0.200	1.127	0.043	0.123	6.368	—	Strong
CT-6	Strain/Condition	25% LB	3	0.244	1.404	0.043	0.123	6.364	—	Strong
SS6 D7	Strain/Condition	100% LB	3	0.754	1.713	0.051	0.149	2.224	—	Strong
CT-7	Strain/Condition	100% LB	3	1.099	1.538	0.051	0.149	1.325	—	Strong
FRP 6	Strain/Condition	25% LB	3	0.281	0.420	0.043	0.123	1.245	—	Strong
CT-6	Strain/Condition	100% LB	3	1.128	1.488	0.051	0.149	1.243	—	Strong
CT-7	Strain/Condition	25% LB	3	0.284	0.404	0.043	0.123	1.164	—	Strong
FRP 6	Strain/Condition	100% LB	3	1.224	1.316	0.051	0.149	0.995	—	Moderate
CTCI SS-3 D7	Strain/Condition	100% LB	3	1.548	1.334	0.051	0.149	0.792	—	Moderate
CTCI SS-3 D7	Strain/Condition	25% LB	3	0.821	0.392	0.043	0.123	0.346	—	Non-biofilm
CTCI-5 D7	Strain/Condition	100% LB	3	1.347	0.585	0.051	0.149	0.337	—	Non-biofilm
CTCI FRP-2 D7	Strain/Condition	100% LB	3	1.330	0.569	0.051	0.149	0.328	—	Non-biofilm
CTCI-1 D7	Strain/Condition	25% LB	3	0.467	0.257	0.043	0.123	0.315	—	Non-biofilm
CTCI SS-2 D7	Strain/Condition	100% LB	3	1.370	0.493	0.051	0.149	0.261	—	Non-biofilm
CTCI-5 D7	Strain/Condition	25% LB	3	0.750	0.304	0.043	0.123	0.256	—	Non-biofilm
CTCI-3 D7	Strain/Condition	100% LB	3	1.409	0.483	0.051	0.149	0.246	—	Non-biofilm
CTCI-2 D7	Strain/Condition	25% LB	3	0.777	0.297	0.043	0.123	0.236	—	Non-biofilm
CTCI FRP-5 D7	Strain/Condition	25% LB	3	0.298	0.183	0.043	0.123	0.234	—	Non-biofilm
CTCI-2 D7	Strain/Condition	100% LB	3	1.306	0.404	0.051	0.149	0.203	—	Non-biofilm



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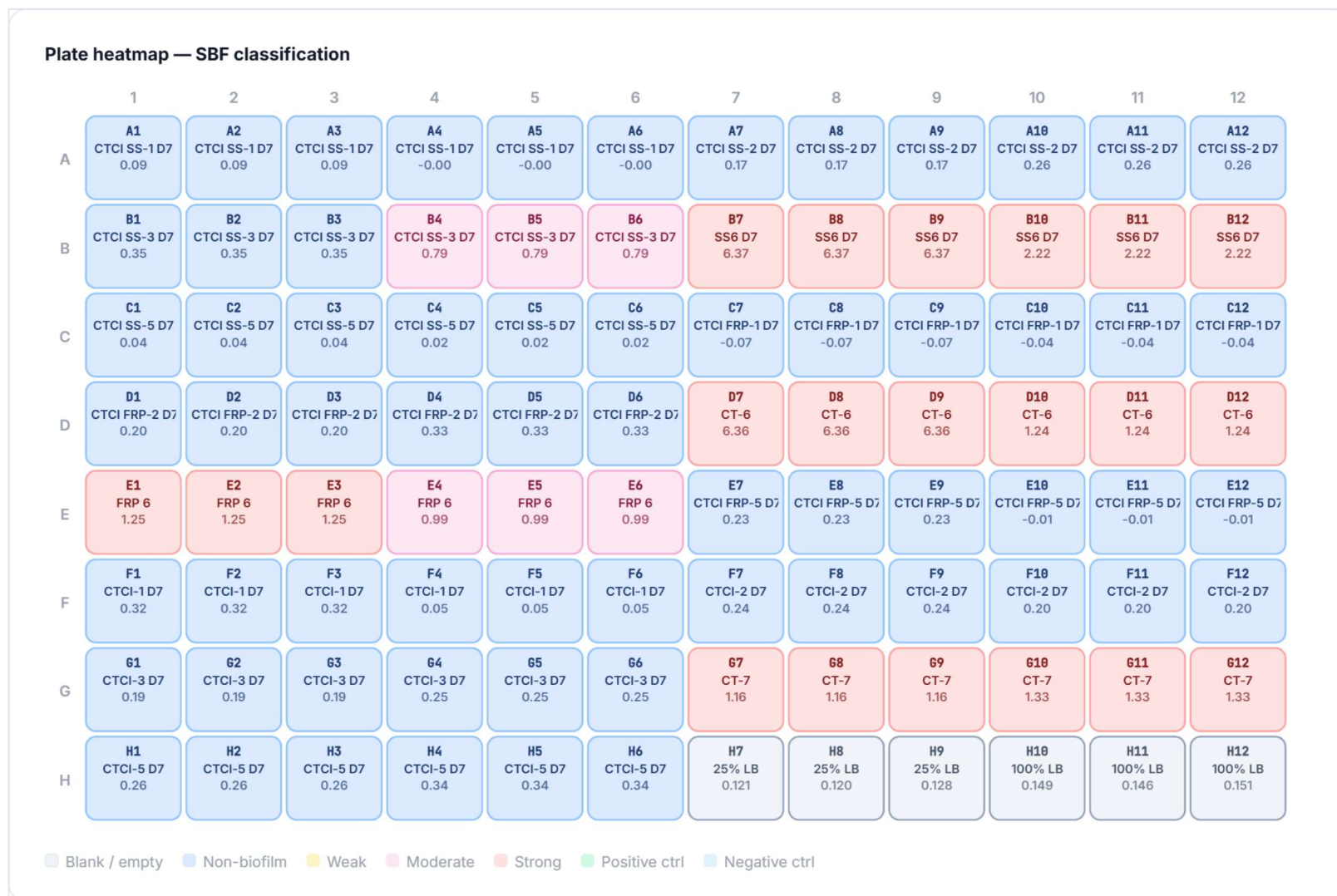
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Strain/Condition / ID	Type	Media	n	Mean OD600	Mean OD570	Blank OD600	Blank OD570	SBF Index	% Inhib.	Classification
CTCI FRP-2 D7	Strain/Condition	25% LB	3	0.813	0.273	0.043	0.123	0.195	—	Non-biofilm
CTCI-3 D7	Strain/Condition	25% LB	3	0.774	0.264	0.043	0.123	0.193	—	Non-biofilm
CTCI SS-2 D7	Strain/Condition	25% LB	3	0.805	0.255	0.043	0.123	0.173	—	Non-biofilm
CTCI SS-1 D7	Strain/Condition	25% LB	3	0.111	0.129	0.043	0.123	0.090	—	Non-biofilm
CTCI-1 D7	Strain/Condition	100% LB	3	0.329	0.163	0.051	0.149	0.049	—	Non-biofilm
CTCI SS-5 D7	Strain/Condition	25% LB	3	0.376	0.138	0.043	0.123	0.044	—	Non-biofilm
CTCI SS-5 D7	Strain/Condition	100% LB	3	0.443	0.158	0.051	0.149	0.023	—	Non-biofilm
CTCI SS-1 D7	Strain/Condition	100% LB	3	0.597	0.147	0.051	0.149	-0.003	—	Non-biofilm
CTCI FRP-5 D7	Strain/Condition	100% LB	3	0.628	0.144	0.051	0.149	-0.008	—	Non-biofilm
CTCI FRP-1 D7	Strain/Condition	100% LB	3	0.578	0.129	0.051	0.149	-0.037	—	Non-biofilm
CTCI FRP-1 D7	Strain/Condition	25% LB	3	0.155	0.116	0.043	0.123	-0.065	—	Non-biofilm

Table 1. Per-strain/condition results, blank- and media-matched. n = number of replicate wells averaged.



6 Plate heatmap — Figure 2





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Figure 2. 96-well plate layout coloured by classification, with per-well SBF index shown.